Verifiable Safe Eviction of Intruding Drone Swarms from Sensitive Areas under Uncertain Tracking and Partial Mitigation Response

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Abstract—An intruding drone swarm creates two coupled counter-UAS problems: move responsive vehicles out of a protected region without creating a collision, and avoid declaring success when tracking is noisy or some vehicles ignore mitigation. We present VERGE-CUAS, a defensive, position-domain framework that joins risk-aware exit assignment, online estimation of per-drone mitigation response, a soft-constrained control-barrier-function quadratic program, and an uncertainty-aware dwell certificate. The certificate propagates position noise into radial-velocity uncertainty and retains enough stand-off for non-reentry even if the mitigation link subsequently drops. A predeclared Monte Carlo study contains 4,000 seeded trials across four methods and five stress scenarios. In 600 proposed-method trials whose members were responsive—including high tracking uncertainty and 35% packet loss—all 600 achieved certified clearance with no core breach, responsive-involved collision, or false clearance (one-sided implication: the 95% Wilson lower bound for each 200/200 scenario is 98.1%). With two of eight or three of sixteen resistant members, VERGE-CUAS certified every responsive subset, identified resistant members at essentially 100%, and correctly returned partial clearance rather than an unsafe full-success claim. These results are simulation evidence, not a field authorization or hardware safety certificate. The implementation intentionally abstracts any authorized mitigation mechanism as a bounded navigation effect and provides no radio-frequency waveform or transmission procedure.

Index Terms—counter-UAS, drone swarm, control barrier function, safe eviction, uncertainty, partial response, clearance certificate

I. Introduction

Low-cost unmanned aerial vehicles can approach a sensitive site as a coordinated group, while the defender must act under incomplete state estimates and uncertain actuation authority. The safety objective is not merely to change a displayed position. It is to produce physical outward motion, preserve separation, avoid modeled ground hazards, and know when the protected region is genuinely clear. Controlled-chamber experiments have shown that commercial UAV motion can sometimes be influenced through navigation deception, but also that takeover is platform-dependent and operationally difficult [1]. Wide-area spoofing research further shows that multiple receivers can be affected coherently [2]. Those results motivate a mechanism-independent control question: given only an authorized bounded navigation effect, what command should be requested, and what evidence is sufficient to declare safe eviction?

Prior directional-expelling work analytically redirects leaderless formations [3]; a recent dual-leader study includes a five-quadrotor experiment [4]. Uncertainty-aware disruption work instead selects nodes whose removal fragments a threatening communication graph [5]. These are important but leave a distinct systems gap: simultaneous protected-zone recovery, inter-vehicle and obstacle constraints, heterogeneous or zero response, noisy tracking, lossy delivery, and a conservative success certificate.

This paper makes four contributions:

1) a bounded state-domain model that separates a defender's requested navigation effect from each vehicle's unknown realized response;
2) a unique-exit assignment minimizing path-integrated ground risk, obstacle proximity, path length, and heading change;
3) an adaptive safety quadratic program (QP) combining finite-time zone recovery with pairwise and obstacle barrier constraints; and
4) a dwell-and-non-reentry certificate plus an honest outcome taxonomy that distinguishes full clearance, partial clearance, safety-violating clearance, and failure.

To the best of our bounded literature search, the novelty is this integration and evaluation rather than any individual primitive. We do not claim that drone eviction, state deception, or control barrier functions are individually new.

II. Related Work and Research Gap

GPS/UAV security experiments focus on feasibility and receiver behavior [1], [2]. Directional expelling and eviction establish how state deception can redirect particular formation models [3], [4]. Graph disruption prioritizes which swarm members should be removed under payload and connectivity uncertainty [5]. In contrast, our dependent variables are protected-zone breach, physical separation, responsive-subset clearance, ground exposure, and false success.

Control barrier functions (CBFs) encode safety constraints as forward-invariance conditions and can be en-

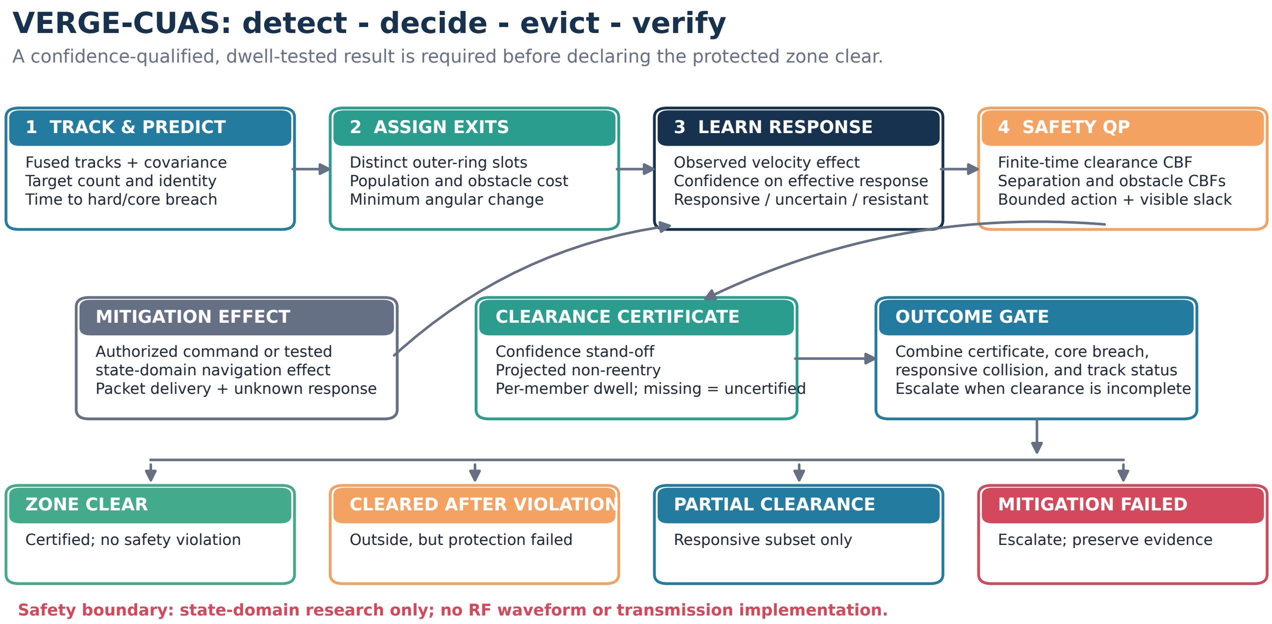

Fig. 1. VERGE-CUAS processing loop. Tracking uncertainty and measured response affect both the safety controller and the independent clearance decision; a resistant member is reported, not silently counted as cleared.

TABLE I
Closest-work capability comparison

Work	Swarm	Evict	Sep.	Uncertain	Cert.
GPS takeover [1]	–	✓	–	empirical	–
False-deviation expelling [3]	✓	✓	–	–	–
Dual-leader eviction [4]	✓	✓	–	–	–
Graph disruption [5]	✓	partial	–	✓	–
CBF geofencing [7]	–	–	✓	robust	barrier
VERGE-CUAS (this work)	✓	✓	✓	✓	dwell+projection

forced using a QP [6]. Hardware geofencing demonstrates that CBF safety filters can operate onboard racing drones [7], while recent work studies finite-time recovery to a safe set under input limits [8]. We use this theory as a controller component, but add uncertain per-agent effectiveness, soft feasibility, exit assignment, and a statistical clearance layer. Table I summarizes the gap.

III. Problem Formulation

A. Protected geometry and threat model

We study a two-dimensional local tangent plane; altitude deconfliction is left to future work. The sensitive-area core and required clearance sets are disks of radii R_{core} and R_{clr}, with $R_{\text{core}} < R_{\text{clr}}$. A tracker supplies associated measurements $\mathbf{y}_{i,k} = \mathbf{p}_{i,k} + \boldsymbol{\nu}_{i,k}$, where $\boldsymbol{\nu}_{i,k} \sim \mathcal{N}(\mathbf{0}, \sigma_{p,i}^2 I)$. The defender does not compromise flight software and does not assume a particular radio implementation. It requests a bounded navigation-domain effect $\mathbf{u}_{i,k}$ through an abstract, authorized interface.

The simulator evolves

$$\mathbf{p}_{i,k+1} = \mathbf{p}_{i,k} + \Delta t \left(\mathbf{v}_{i,k}^0 + a_i b_{i,k} \mathbf{u}_{i,k} + \mathbf{w}_{i,k} \right), \qquad (1)$$

where $\mathbf{v}_{i,k}^0$ is the nominal intruder velocity, $a_i \in [0, 1]$ is unknown response effectiveness, $b_{i,k} \in \{0, 1\}$ is packet delivery, and $\mathbf{w}_{i,k}$ is process noise. Commands obey $\|\mathbf{u}_{i,k}\|_2 \leq u_{\max}$. A member is operationally resistant if its learned response upper confidence bound falls below $a_{\min}$.

The defender may not be able to clear a member with $a_i \approx 0$. Therefore our achievable requirement is: clear every responsive member, prevent responsive-involved collisions and core breach when all members are responsive, and never label a mixed swarm fully clear while an uncleared member remains.

B. Outcome definitions

Let $c_i \in \{0, 1\}$ denote the final certificate. We report:

- zone clear: all $c_i = 1$, with no core breach, responsive collision, or false certificate;
- cleared after safety violation: all certified, but a safety condition was violated;
- partial clearance: at least one but not all members certified; and
- mitigation failed: no member certified by the horizon.

This taxonomy prevents a high responsive-subset success rate from being misreported as complete site protection.

IV. The VERGE-CUAS Method

Figure 1 gives the full loop.

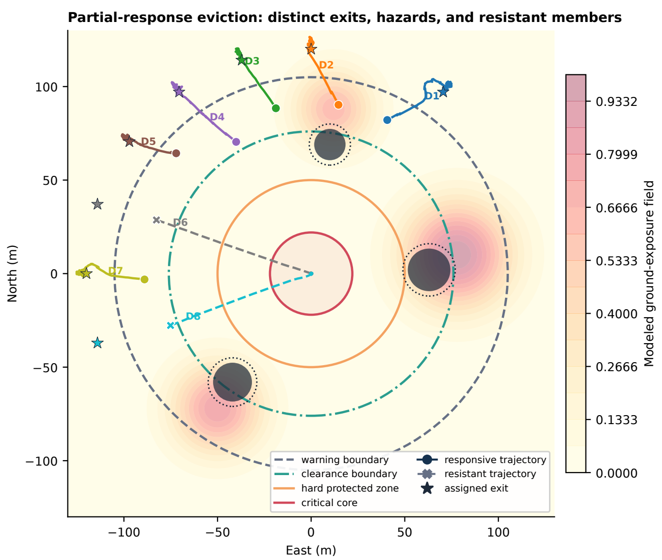

Fig. 2. Evaluation geometry and representative mixed-response behavior. Unique exits separate responsive trajectories; resistant members can continue inward and force a partial-clearance outcome.

A. Risk-aware unique exit assignment

Candidate exits $\mathbf{e}_j$ lie on a radius-R_e ring beyond the warning boundary. For drone i and slot j, 21 points sample the straight reference segment. Its assignment cost is

$$C_{ij} = L_{ij} + \lambda_r L_{ij} \int_0^1 \rho(\boldsymbol{\ell}_{ij}(s))ds + \lambda_o \Psi_{ij} + \lambda_\theta (1 - \hat{\mathbf{p}}_i^T \hat{\mathbf{e}}_j), \tag{2}$$

where ρ is a smooth exposure field, Ψ penalizes small or negative obstacle clearance, and L_{ij} is path length. The Hungarian algorithm [9] finds the minimum-cost one-to-one assignment. Unlike a common exit, uniqueness does not intentionally funnel the swarm into one point.

B. Online response estimation

The measured control effect is $\mathbf{z}_{i,k} = \hat{\mathbf{v}}_{i,k} - \mathbf{v}_{i,k}^0$. When $\|\mathbf{u}_{i,k}\|^2 > 1$, a scalar observation is

$$\tilde{a}_{i,k} = \mathrm{clip}_{[0,1.2]} \frac{\mathbf{u}_{i,k}^T \mathbf{z}_{i,k}}{\|\mathbf{u}_{i,k}\|_2^2}, \quad \hat{a}_{i,k+1} = (1-\eta)\hat{a}_{i,k} + \eta \tilde{a}_{i,k}. \tag{3}$$

An information score I_i accumulates command excitation, producing $\sigma_{a,i} = (\sigma_{a,0}^{-2} + I_i)^{-1/2}$. A member is flagged resistant when $\hat{a}_i + 1.96\sigma_{a,i} < a_{\min}$. The controller then stops relying on that member to execute avoidance and allocates responsive-neighbor motion instead.

C. Safety QP

The nominal exit-tracking request is

$$\mathbf{u}_i^{\mathrm{ref}} = \mathrm{sat}_{u_{\max}} \left(\frac{v_d \widehat{(\mathbf{e}_i - \mathbf{y}_i)} - \mathbf{v}_i^0}{\max(\hat{a}_i, a_{\min})} \right). \tag{4}$$

We define a clearance recovery barrier $h_i^z = \|\mathbf{y}_i\|^2 - R_{\mathrm{clr}}^2$, pairwise barrier $h_{ij}^s = \|\mathbf{y}_i - \mathbf{y}_j\|^2 - d_{\min}^2$, and analogous circular-obstacle barriers h_{iq}^o. The zone constraint uses a finite-time term:

$$\dot{h}_i^z + \kappa_z \,\mathrm{sgn}(h_i^z)|h_i^z|^\gamma + s_i^z \geq 0, \quad 0 < \gamma < 1. \tag{5}$$

Separation and obstacle constraints use $\dot{h} + \kappa h + s \geq 0$. At each 0.2 s step, we solve

$$\min_{\mathbf{u},\mathbf{s}\geq 0} \ \frac{1}{2}\|\mathbf{u} - \mathbf{u}^{\mathrm{ref}}\|_2^2 + \frac{1}{2}\sum_r w_r \|\mathbf{s}^r\|_2^2, \tag{6}$$

$$\text{s.t.} \quad (5),\ \text{separation/obstacle CBFs},\ \|\mathbf{u}_i\|_2 \leq u_{\max}. \tag{7}$$

The norm ball is implemented as a 12-sided inscribed polygon. Slack priorities are $w_s > w_o > w_z$, so feasibility is retained while collisions receive the largest penalty. OSQP solves the sparse QP [10]; a bounded reference command is the recorded fallback if the solver does not return an acceptable solution.

D. Uncertainty-aware clearance certificate

A declaration based only on $\|\mathbf{y}_i\| > R_{\mathrm{clr}}$ is unsafe: differencing noisy positions amplifies velocity error, and a subsequent packet drop may restore inward motion. With exponential velocity-filter gain q, we propagate

$$\sigma_v = \frac{\sqrt{2}\sigma_p}{\Delta t}\sqrt{\frac{q}{2-q}}, \quad \underline{r}_i = \|\mathbf{y}_i\| - z\sigma_{p,i}, \tag{8}$$

and a lower radial-velocity bound

$$\underline{v}_{r,i} = \hat{\mathbf{r}}_i^T \hat{\mathbf{v}}_i - z\sigma_v. \tag{9}$$

To remain conservative after a link change, the inward-speed envelope is

$$v_i^{\mathrm{in}} = \max\{-\underline{v}_{r,i},\ v_{\max}^0 + z\sigma_w\}. \tag{10}$$

Drone i is certified only if (i) $\underline{r}_i \geq R_{\mathrm{clr}}$ continuously for $m = T_h/\Delta t$ samples and (ii)

$$\underline{r}_i - T_h v_i^{\mathrm{in}} \geq R_{\mathrm{clr}}. \tag{11}$$

We use $z = 2.576$ and $T_h = 3$ s. Certification is absorbing for scoring, but a real implementation should continue tracking and revoke site status after a new incursion.

V. Experimental Method

A. Predeclared scenarios and baselines

Table II lists the fixed configuration. Each method–scenario cell contains 200 independent, deterministically seeded trials, for 4,000 main trials. Initial radii are 80–97 m; inward speeds are 3–5 m/s. The five scenarios are: all responsive; two of eight resistant; tracking standard deviation 1.2 m; 35% packet loss; and a sixteen-member mixed swarm with three resistant members. The latter two mixed cases are intentionally impossible to clear completely through a navigation effect alone.

Baselines are no mitigation, proportional control to one common exit, and proportional control to the nearest radial exit. All methods use the same tracker samples and trial seeds within a scenario. Three ablations remove

TABLE II
Primary simulation configuration

Quantity	Value
Step / horizon	0.2 s / 50 s
Core / hard / clearance radius	22 / 50 / 76 m
Warning / exit-slot radius	105 / 120 m
Minimum separation / collision	8 / 2 m
Command limit / desired speed	14 / 9 m/s
Certificate confidence / dwell	2.576 / 3 s
Primary swarm / large swarm	8 / 16 members
Trials	200 per method–scenario cell

TABLE III
VERGE-CUAS main outcomes (200 trials per row)

Scenario	Safe %	Any %	Core %	Resp. col. %	False %	$\tilde{t}_c$ s
All responsive	100	100	0	0	0	3.0
High uncertainty	100	100	0	0	0	4.2
35% packet loss	100	100	0	0	0	4.2
2/8 resistant	0	100	100	0	0	–
3/16 resistant	0	100	100	0	0	–

response adaptation, risk-aware assignment, or the dwell certificate. Sensitivity sweeps vary command limit, confidence multiplier, and hold time.

B. Metrics and statistical analysis

The primary metric is safe clearance: complete certificate and no core breach, responsive-involved collision, or false clearance. Secondary metrics include partial-or-complete clearance, clearance time, minimum separation, command-energy proxy $\sum_k \|\mathbf{u}_k\|^2 \Delta t$, integrated ground risk, resistant identification, re-entry, and QP fallback steps. We distinguish unavoidable resistant–resistant collisions from collisions involving at least one responsive member. Binary 95% intervals use the Wilson score method [11]; clearance-time intervals use 4,000 seeded bootstrap resamples of the median.

VI. Results

A. Main outcomes

Table III reports proposed-method outcomes. All 600 fully responsive trials safely cleared. The median was 3.0 s nominally, 4.2 s with high uncertainty, and 4.2 s under packet loss. No proposed-method trial in those scenarios breached the core, collided, or declared a false clearance. For 200/200 successes, the two-sided 95% Wilson interval is [98.1%, 100%]; zero observed failures should not be interpreted as proof of zero real-world failure probability.

In both mixed cases, the responsive subset was certified in every trial and no responsive-involved collision occurred. Resistant members continued inward, yielding a core breach and resistant-only collision in every trial; therefore complete safe clearance was correctly 0%. This is not a controller success for full site protection. It is evidence that the framework identifies the boundary of its authority and refuses a false positive.

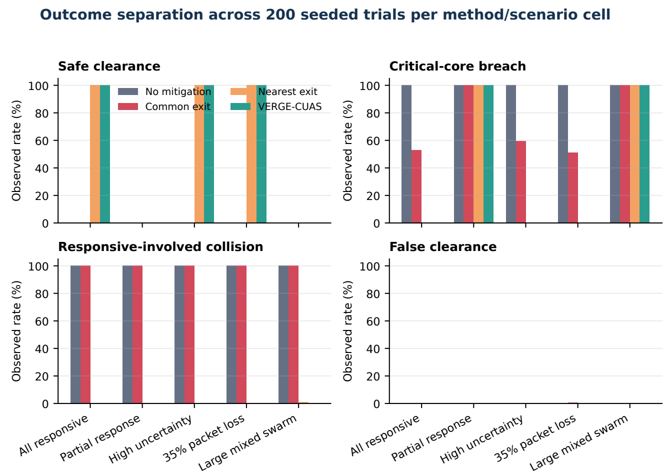

Fig. 3. Outcome and safety comparison across methods and scenarios. Complete clearance alone is misleading for the common-exit controller because its paths collide.

The common-exit baseline eventually certified all members in 97–100% of fully responsive cases but caused a responsive collision in every trial and core breach in 51–59.5%, so safe clearance was 0%. Nearest radial exits matched full clearance in the three responsive scenarios but caused two responsive-involved collisions among 200 large mixed-swarm trials and had no resistant identification mechanism. VERGE-CUAS used 47% less median command energy than nearest-exit control in the large mixed scenario (67,418 versus 127,057 in simulator units) because identified resistant members were not treated as reliable actuators.

B. Adaptive response and certification

Figure 4 shows a representative mixed-response trial: responsive estimates remain above the threshold, resistant upper confidence bounds fall below it, and certificates accrue only after conservative boundary dwell. Across the two mixed scenarios, mean resistant identification was 100% (99.8–100% before rounding in individual batches), with zero responsive false-resistant classifications in the main run. The 4,000-trial study recorded 17 QP fallback steps for VERGE-CUAS out of at most 1,000,000 update opportunities; fallback use was logged rather than removed from scoring.

Removing the certificate and accepting one sample produced false-clearance rates of 26% nominally, 28% under high tracking noise, 75% with two resistant members, and 94% in the large mixed swarm. This ablation is the clearest evidence that movement generation and success verification are separate problems. Fixed response assumptions retained responsive-scenario safety but lost resistant classification and increased median command energy by 14.7% (8-drone mixed) and 21.3% (16-drone mixed). Risk-integrated assignment reduced median exposure relative to nearest radial exits by 4.9% in the

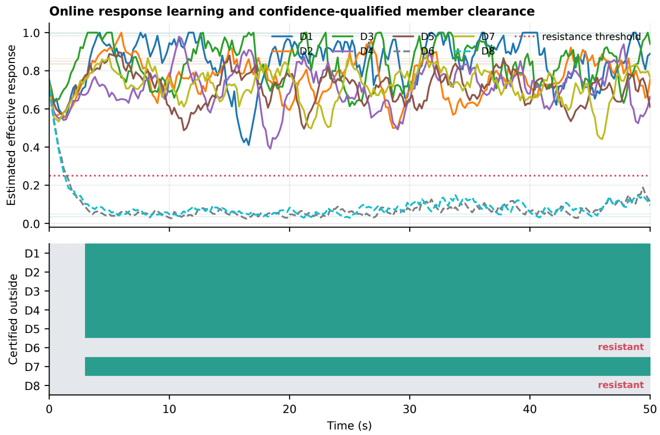

Fig. 4. Representative online response estimates and clearance events. The certificate lags geometric crossing by design.

partial-response ablation (10.26 versus 10.79 units) but increased it by 13.8% for the large mixed swarm (23.88 versus 20.97). Thus, unique assignment remains useful for deconfliction, but the present static path-risk surrogate does not guarantee lower closed-loop exposure.

C. Sensitivity and reproducibility

The 1,500-trial sensitivity study is retained as a vector figure and raw table in the artifact. All command limits from 8–16 m/s yielded 100/100 safe clearances, but the 8 m/s setting required 206 recorded QP fallbacks and had a 6.6 s median versus 3.0 s at 12 m/s or above. Raising z from 0 to 3.291 preserved 100/100 observed safe clearance but increased the median from 3.0 to 7.8 s. Dwell times through 5 s preserved 100/100, while 7 s produced only 23/100 completion by the 50 s horizon (the completed-trial median was 38.6 s). The selected 3 s dwell is therefore a tested operating point, not a universal optimum.

All source, fixed seeds, raw trial rows, manifests, interval calculations, vector figures, and unit tests accompany the manuscript. The complete study comprises 7,100 trials: 4,000 main, 1,600 ablation, and 1,500 sensitivity trials. The test suite covers signed geometry, one-to-one assignment, estimator convergence, bounded QP output, confidence-and-dwell behavior, deterministic replay, and an end-to-end false-clearance smoke case.

VII. Safety, Legal Scope, and Limitations

This work is a defensive simulation. It neither implements nor instructs radio-frequency interference, satellite-signal generation, protocol exploitation, destructive capture, or outdoor transmission. India's Department of Telecommunications distinguishes non-radiating indoor experimentation from radiating indoor/outdoor trials and provides a licensing process [12]. Any hardware work must therefore use an authorized facility, spectrum approval, institutional risk review, containment, geofencing, a pilot-in-command, emergency termination, and owned/consenting aircraft. The paper's state-domain input should be realized first through simulation, hardware-in-the-loop, or a wired/shielded interface.

The principal limitations are substantial. The plant is a planar point mass, not a flight-controller or aerodynamics model. Track association is assumed correct; clutter, identity swaps, occlusion, altitude, wind fields, and non-Gaussian bias are not modeled. The response estimator is scalar and cannot represent direction-dependent or delayed effects. CBF slack means the optimizer remains numerically feasible but does not create a theorem that an uncontrollable member will be safe. The Gaussian confidence calculation is model-conditional, and the Monte Carlo results cannot establish certification for untested platforms. Finally, a resistant intruder is not evicted by this navigation-only abstraction; a real defender requires a separate lawful escalation and site response.

VIII. Conclusion

VERGE-CUAS reframes counter-UAS eviction as both a constrained control problem and a verifiable decision problem. The study shows that unique exits and barrier constraints prevent the funneling failures of a common target, response learning avoids treating resistant members as controllable, and a dwell-plus-worst-inward-motion certificate eliminates false success in the evaluated trials. The strongest result is conditional: all responsive stress cases cleared safely in simulation. The equally important negative result is that navigation-domain influence cannot protect the core from nonresponsive members; the correct output is partial clearance, not an inflated success claim. Future work should validate the model in hardware-in-the-loop, add 3-D reachable sets and track-management faults, and integrate a lawful escalation layer for resistant vehicles.

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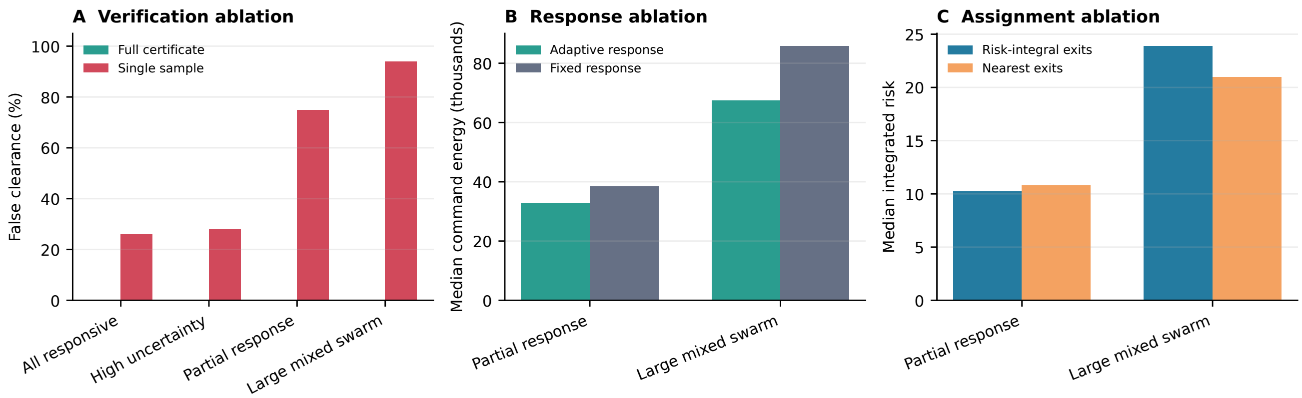

Fig. 5. Paired component ablations (100 trials per method–scenario cell). The certificate has the dominant effect on truthful outcomes; adaptation reduces wasted command energy; the static assignment cost has scenario-dependent exposure.

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